IRREGULAR-SINGULAR KZB REGULARITY FROM BOUNDARY FORMAL TYPE AND BOUNDARY-FORMAL-TYPE COMPOSITION

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Abstract. The modular operad $\mathcal{O}^{A_\infty\text{-ch}}$ governing A_∞-algebras in E_1-chiral algebras has five axioms. Ordinary nodal KZB degeneration is Fuchsian: it has slope 0 and no Stokes sector. Irregular Stokes-sector composition enters only after a compatible good boundary formal type has been supplied on every stable-graph boundary stratum. Edgewise Stokes data are not sufficient: two-edge corners require a mixed-channel flatness condition. Under that hypothesis, the Jimbo–Miwa–Ueno framework supplies Stokes-sector composition of flat sections. No general $\mathcal{W}_N$ Poincaré-rank or Stokes-ray count is asserted here.

1. The composition axiom

The modular operad $\mathcal{O}^{A_\infty\text{-ch}}$ has five axioms. Four have the following proved or datum-relative scopes:

(i) genus-0 product decomposition,
(ii) $\pi_1(\Sigma_g)$ well-definedness (KZB flatness),
(iii) $\mathrm{Aut}(\Gamma)$-equivariance on the braided annular quantum-group branch (quasi-triangularity, Yang–Baxter braid coherence, pure-braid descent, and inverse orientation),
(iv) unitality for unit-normalised annular sewing (vacuum/counit normalisation of the diagonal seam).

The fifth axiom, *composition associativity under clutching*, is unconditional at genus 0 for every non-critical level and at all genera for integrable levels (Kazhdan–Lusztig and its KZB extension by Tsuchiya–Ueno–Yamada, Beilinson–Drinfeld). In an irregular boundary sector the composition law is the following Stokes-sector theorem.

Theorem 1.1 (Irregular-singular KZB regularity). *Let A be a chirally Koszul algebra at non-critical level k. Assume that, at every boundary point $p \in \partial\overline{\mathcal{M}}_{g,n}$ of every stable-graph stratum, including two-edge corners, either the ordinary nodal Fuchsian type is used or a compatible good irregular formal type is supplied; in the irregular case assume also residue-gap nonresonance and vanishing mixed-channel Stokes curvature. Then ∇^{KZB} has:*

(i) *slope 0 in the ordinary nodal case, and in the supplied irregular case rational slopes with denominators given by the finite ramification indices in $q_e = t_e^{N_e}$;*
(ii) *after finite ramification, formal type $\Lambda_p(t) = \sum_{m=1}^{d} A_m t^{-m}$, where d is pole order in the ramified coordinate, determines the Stokes directions and Stokes groups;*
(iii) *sectorial asymptotic expansions of flat sections, canonical in each Stokes sector, with Deligne–Malgrange Stokes matrices $\{S_\sigma\}$ for transitions; these matrices are analytic Stokes data of the connection, not determined by the formal type;*
(iv) *Jimbo–Miwa functional relations for $\{S_\sigma\}$ along admissible families preserving the formal type, Stokes directions, zero-curvature equation, and stable graph;*
(v) *well-defined clutching composition at p via Stokes-sector composition of flat sections, coherent under further clutching.*

This is Vol. II Theorem ?? in Chapter ??. Jimbo–Miwa–Ueno isomonodromic technology, Boalch's Stokes-space formalism, and the Kohno–Felder elliptic KZB connection at genus ≥ 1 are the external inputs. The boundary formal-type compatibility must be supplied beyond edgewise Stokes classification.

2. KZB and its singularities

The KZB connection on $\mathrm{Conf}_n(\Sigma_g)$ reads

$$\nabla^{\mathrm{KZB}} \;=\; d \;-\; \sum_{i<j} \Omega_{ij}\, \omega_{ij} \;-\; \sum_{i=1}^{n} \sum_{\alpha=1}^{g} H_i^{\alpha}\, d\Omega_{\alpha} \;+\; \cdots,$$

with $\Omega_{ij}/(k+h^{\vee})$ the KZB connection residue in the standard normalisation, ω_{ij} the Szegő kernel on Σ_g, Ω_{α} the period matrix, and the ellipsis denoting Eisenstein-type corrections at genus ≥ 2. The bar collision-residue convention $k\,\Omega_{\mathrm{tr}}/z$ is a different normalisation. The connection is flat in its non-critical domain.

2.1. Genus-0 regularity.

Proposition 2.1 (Genus-0 regularity). *The KZ connection form $\omega = \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j)$ is manifestly logarithmic; singularities at boundary divisors of $\overline{\mathcal{M}}_{0,n}$ are regular for every k.*

2.2. Integrable level.

Proposition 2.2 (Integrable-level regularity). *At $k \in \mathbb{Z}_{\geq 0}$ the KZB connection extends with regular singularities to $\overline{\mathcal{M}}_{g,n}$ for all g.*

Kazhdan–Lusztig plus TUY plus Beilinson–Drinfeld. The condition that made this work was integer denominators and semisimple fusion.

2.3. **Formal-type nonresonance.** Nonintegrality of k is not a Stokes condition. At an ordinary nodal boundary divisor the KZB connection is logarithmic and regular singular. Resonance is controlled by residue eigenvalue gaps divided by $k+h^{\vee}$, and irregularity appears only after a good formal type with higher polar terms has been chosen. Theorem 1.1 replaces Kazhdan–Lusztig regular-singular parallel transport only on that supplied irregular boundary sector.

3. The Jimbo–Miwa framework

3.0.0.1. Newton polygon at a boundary point. At a stable-graph stratum with node p and transverse coordinate ${}_p$, the connection near p reads

$$\nabla^{\mathrm{KZB}} \;=\; d \;-\; \frac{A_1}{{}_p}\, d\;{}_p \;-\; \sum_{j\geq 2} \frac{A_j}{{}_p^{\,j}}\, d\;{}_p \;-\; \omega_{\mathrm{reg}},$$

with A_j level-dependent endomorphisms of the tensor product of evaluation modules. The Newton polygon is the lower convex hull of $\{(j, \mathrm{ord}(A_j))\}$; its slopes are the irregularity slopes. The residue A_1 is the quadratic Casimir $\Omega_{\mathrm{node}}/(k+h^{\vee})$. A simple node has pole order one, is regular singular, and has irregular slope 0. Higher terms A_j are not automatic consequences of the nodal KZB connection; they are part of the supplied good irregular formal type, for example after a ramification $q = t^N$.

3.0.0.2. Stokes structure. The formal type determines the Stokes directions and Stokes groups. Nonresonance is the condition that residue eigenvalue gaps divided by $k+h^{\vee}$ avoid integers; nonintegrality of k alone is neither necessary nor sufficient. The matrices $S_{\sigma} \in \mathrm{Sto}_{\sigma}(\Lambda)$ are analytic Stokes data of the connection, locally constant under isomonodromic deformation.

3.0.0.3. Jimbo–Miwa functional relations. For an admissible family preserving the formal type, Stokes directions, zero-curvature equation, and stable graph, the isomonodromic deformation is governed by Jimbo–Miwa (1981); the Stokes matrices $\{S_\sigma\}$ are locally constant on moduli up to Stokes wall-crossings (Berk–Nevins–Roberts). The forgetful map $\overline{\mathcal{M}}_{g,n} \to \overline{\mathcal{M}}_{g,n-1}$ is not automatically such a family.
3.0.0.4. Clutching composition. Near a node the clutching map $\mu_p\colon \mathfrak{g}^{\mathcal{A}}_{\mathrm{mod}}(g_1, n_1) \otimes \mathfrak{g}^{\mathcal{A}}_{\mathrm{mod}}(g_2, n_2) \to \mathfrak{g}^{\mathcal{A}}_{\mathrm{mod}}(g_1 + g_2, n_1 + n_2 - 2)$ is the Stokes-sector composition of flat sections on either side of the node. Associativity under further clutching reduces to the Jimbo–Miwa Schlesinger system.

4. The $\mathcal{W}_N$ depth datum

Proposition 4.1 ($\mathcal{W}_N$ depth is independent of a KZB Stokes count)**.** *The integer $2N - 2$ measures the depth gap between the Virasoro generator of weight 2 and the highest principal generator of $\mathcal{W}_N$. The Poincaré rank of the KZB connection at a boundary divisor is separate data. A KZB Stokes count requires an explicit irregular normal form and a Stokes-sector analysis for that normal form.*

Thus this standalone uses the $\mathcal{W}_N$ depth only as a representation-theoretic datum. It does not assert a universal Stokes ray formula.

5. Consequences

Under the boundary formal-type hypothesis, irregular composition is Stokes-sector composition. Ordinary nodal composition remains regular-singular monodromy. On any overlap where both descriptions are defined, the sectorial construction restricts to the regular-singular one. The consequences used by the modular surface are therefore conditional on Theorem 1.1:

- *Curved-Dunn H^2-vanishing:* Stokes-sector composition does not prove curved-Dunn vanishing; the vanishing is an algebraic input from modular-bootstrap acyclicity plus the degree-2 comparison map.
- *Modular-operad composition:* the irregular composition axiom is supplied by the conditional Stokes-sector theorem after the formal type and mixed-corner flatness are specified.
- *R-twisted clutching:* R-twisted clutching uses the algebraic curved-Dunn comparison and the analytic Stokes-sector composition as separate inputs.
- *Composition axiom:* the clutching composition law in the irregular formal-type sector is Stokes-sector composition.

6. Dependencies

The argument depends on:

- Vol. II Theorem **??** (main input);
- Vol. II Proposition **??** (conditional degree-2 comparison transporting modular-bootstrap acyclicity to curved-Dunn acyclicity);
- Vol. II Proposition **??** (genus-1 twisted tensor criterion);
- external: Jimbo–Miwa–Ueno 1981; Boalch 2001/2014; Felder 1994 (Kohno–Felder elliptic KZB).